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Pivotal Trial Validating Usability and Visualization Performance of Home OCT in Neovascular Age-Related Macular Degeneration: Report 1

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Purpose: To validate the usability and visualization performance of the index test of the home OCT system (HOCT) during a pivotal study toward de novo US Food and Drug Administration marketing authorization.

Design: A prospective, 5-week longitudinal, at-home visualization multicenter study with preplanned office visits at week 1 and week 5 and as-needed interim visits.

Participants: The study enrolled adults aged ≥55 years diagnosed with neovascular age-related macular degeneration (nAMD) on anti-VEGF therapy in at least 1 eligible eye and best-corrected visual acuity of 20/320 or better.

Methods: Participants self-installed and imaged daily with the HOCT at home for 5 weeks with 2 or 3 interspersed office visits at 1 and 5 weeks with interim reading center (RC)-triggered visits including a comparator in-office OCT (IO-OCT). Scans with an acceptable quality signal index were independently graded by the RC in a masked manner.

Main Outcome Measures: Ability to self-image at home, positive and negative percent agreement (NPA) in visualization of total hyporeflective spaces (TRO) on HOCT and on IO-OCT.

Results: At home, self-imaging success rate was 96.1% (95% confidence interval [CI]: 92.2%–98.4%). One hundred eighty participants self-imaged the primary and secondary eyes 5426 and 4012 times with a mean (standard deviation) manufacturer signal quality index of 4.40 (1.26) and 4.58 (1.28), respectively. Positive percent agreement was 86.6% (95% CI: 80.4%–92.8%) and NPA was 86.1% (95% CI: 80.4%–91.8%), with nearly all disagreements being minimal.

Conclusions: The target population successfully self-installed and self-imaged at home with image quality comparable to IO-OCT. The findings of the visualization study support the intended use of the system as a tool to monitor TRO at home between routine clinical visits during the management of nAMD.

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Supplemental material available at www.ophthalmologyscience.org.

The combination of intravitreal anti-VEGF agents and in-office OCT (IO-OCT)–guided dosing regimens has transformed neovascular age-related macular degeneration (nAMD) from a progressive, chronic disease with visual loss into a treatable, chronic disease with the potential for maintenance or improvement in vision.[1–5] However, achieving these outcomes can be costly, with therapies for nAMD totaling US $10 billion per year across the entire US health care system.[6] Despite the high intensity and cost of therapy, studies have shown that visual outcomes for patients in the clinical setting remain significantly below those reported in clinical trials, which is often attributed to undertreatment. Undertreatment, in turn, is thought to be partly a result of waning compliance due to suboptimal follow-up and treatment regimens implemented by the health care providers.[7,8] More individualized treatment approaches, especially with longer duration therapies, have the potential to overcome these limitations. However, due to the expected variability among patients, frequent disease monitoring with OCT is still needed to minimize undertreatment. If there was an option to perform such follow-up monitoring using a home OCT (HOCT), rather than in-office visits and OCT, this would greatly reduce the burden on patients, caregivers, and providers while potentially reducing the direct and indirect costs associated with nAMD. However, such an approach requires 2 key components: a device that allows self-acquisition of proper OCT images at home and an artificial intelligence–based algorithm to extract critical disease parameters from the high volume of acquired image data.[9]

Recent studies of a patient using the self-installed and self-operated spectral-domain OCT system from the same

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manufacturer demonstrated highly successful daily self-imaging at home approximately 6 times per week as well as excellent fidelity of the associated automated algorithm to quantify OCT-based disease parameters with more than 90% of scans eligible for analysis.[10–15] However, these studies included a smaller sample size or a retrospective analysis; therefore, they were not designed to formally validate these key capabilities. In agreement with the US Food and Drug Administration (FDA), a combination of a longitudinal study and a cross-sectional study was designed to support respective elements of the following indication for use statement toward a de novo marketing authorization of a new class of devices defined by the FDA as "Home Monitoring Ophthalmic Imaging Device."

The Notal Vision HOCT System is an artificial intelligence–based home use device indicated for visualization of intraretinal and subretinal hyporeflective spaces (SRO) in a 10° by 10° area centered on the point of fixation of eyes diagnosed with nAMD. In addition, it provides segmentation and an estimation of the volume of hyporeflective spaces (HRS). The Notal HOCT device is intended for imaging at home between regularly scheduled clinic assessments and is not intended to be used to make treatment decisions or replace standard-of-care regularly scheduled examinations and clinical testing as needed, including in-office imaging and assessments for changes in vision by an ophthalmologist.

The purpose of the "Home OCT Fluid Visualization Study," the subject matter of Report 1, is to characterize the performance of the Notal Vision HOCT in visualization of biomarkers associated with nAMD by the target population as presented to FDA in support of the label's indication for use statement.

Methods

The "Home OCT Fluid Visualization Agreement Study" was a prospective, multicenter study approved by the Advarra Institutional Review Board and conducted in accordance with the tenets of the Declaration of Helsinki. Written informed consent was obtained from all participants. The clinical trial registration information is publicly available at https://clinicaltrials.gov (NCT04907409). It was conducted at 7 US retina clinics from June 22, 2021, to December 15, 2022.

The HOCT Device

Details of the SCANLY HOCT system (Notal Vision, Inc) have been previously described.[10–14] Briefly, it is an spectral-domain OCT device designed with a compact form factor compared to an office OCT (18 lbs/8 kg, 14 inches/35.6 cm width, 10.6 inches/27 cm depth, 14 inches/35 cm height) for patient's self-operation at home. It uses a fixed horizontal raster scan pattern of 88 B-scans over a 3 mm × 3 mm (3 × 3) area or 10° × 10° field of view with axial resolution of 19 μm and transverse resolution of 35 μm. Patient head positioning and fixation are guided by a proprietary automatic visual and auditory feedback system. During a patient's first use, the device performs an automatic calibration to adjust for each eye's refractive error and axial length. The captured scans are automatically uploaded to the Notal Health Cloud via a built-in cellular modem. Once volume scans are reconstructed, a manufacturer signal quality index (MSI) is calculated that provides an objective and quantifiable representation of image quality and is reported on a scale of 0 to 7. An MSI ≥2 is the recommended level for acceptable image quality. Details of the construction of the MSI and determination of the recommended cutoff were previously described.[11,12] Upon an event of a device exchange for technical reasons at the patient's home, a graphic indicator is displayed in the physician report. Due to the risk of the development of a rapid exudative process, the patients are asked to self-image daily. Prior studies have shown that good adherence to daily regiment can be achieved during HOCT studies and in real-world use of the ForeseeHome (Notal Vision).[11,13,14,16] An explanation for these findings may include the effect of the physician recommendation, the ease of use of the system, and the patient's perspective about the importance of frequent monitoring. The volume scans are analyzed by the Notal OCT Analyzer and are available for the physician's review. Details of the Notal OCT Analyzer have been described elsewhere, and its validation is described in Report number 2.[13,17–19] The output of the system includes a section that allows simultaneously scrolling through all the B-scans at 2 time points of interest among other elements and visualizing HRS as well as any other retinal findings (Fig 1).

Additional notes are about SRO, intraretinal hyporeflective spaces (IRO), and total hyporeflective spaces (TRO).

The terms total retinal fluid, subretinal fluid, and intraretinal fluid are commonly used to refer to HRS observed on B-scan images of OCT, termed here TRO, SRO, and IRO. However, not all HRS on macular OCT imaging represent exudative retinal fluid, and not all exudative fluids will necessarily be hyporeflective; therefore, the terms total retinal fluid, subretinal fluid, and intraretinal fluid are not synonymous with the terms TRO, SRO, and IRO.[20] The definitions used by the reading center (RC) (MERIT CRO) during the study and accordingly the entire report refer to HRS as TRO, SRO, and IRO, which are limited to the observation of images while not making any physical interpretation.

The "Home OCT Fluid Visualization Study" was a prospective, longitudinal study. Adults aged ≥55 years with a prior diagnosis of nAMD in at least 1 eye and best-corrected visual acuity of 20/320 or better were enrolled by their retina physician 1 week before a previously scheduled, routine clinic visit. Those who required anti-VEGF treatment for nAMD in the study eye at the screening visit, those who had any other retinal disease requiring intravitreal therapy, or those with prior HOCT device use experience were excluded; otherwise, the presence of concomitant pathologies was not exclusionary. One nAMD eye was determined to be the primary study eye. For the subjects with both eyes with nAMD, 1 eye was assigned as the primary eye according to a randomization schedule and the other was the secondary eye. The purposes of the study were to evaluate the following: (1) the agreement between comparator IO-OCT and the index test HOCT in the visualization of the target condition of HRS in the central 10° of the macula, as determined by 3 expert graders at a third-party RC, and (2) the ability of participants to successfully self-image with the HOCT device. Reading center graders were masked to each other's determinations, the source device, and the participant ID number, eliminating any possible bias. Since it was possible that the graders would distinguish between the HOCT and IO-OCT scans, a washout period was set to verify that all gradings were performed while isolating between scans of the same eye associated with the same visit. The review order of the scans was randomized, and disagreements between graders were adjudicated. Intraretinal hyporeflective spaces and SRO were graded as present when definite HRS was observed in individual scans and did not include

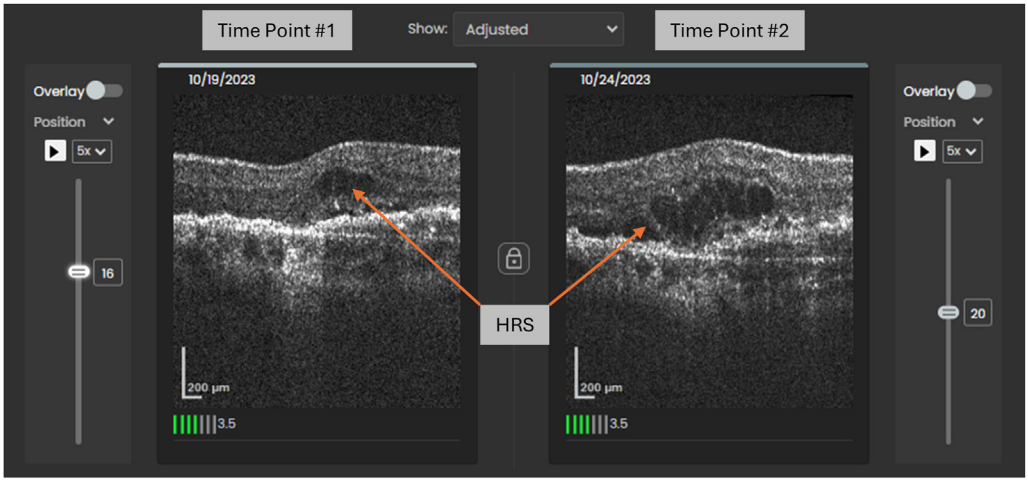


Figure 1. The B-scans visualization module of the home OCT physician report. It allows simultaneous scrolling through all the B-scans at 2 time points of interest and visualization of HRS. HRS = hyporeflective spaces.

quantification or assessment of a change in HRS. Intraretinal hyporeflective spaces was defined as a hyporeflective space above the photoreceptor layer, while SRO was defined as HRS below the photoreceptor layer and above the retinal pigment epithelium. The presence of confounding lesions (epiretinal membrane, macular hole, pseudocysts, outer retinal tubulations, hemorrhage, pigment epithelial detachments, subretinal hyper-reflective material, geographic atrophy, and hyper-reflective retinal spots or foci) was also assessed by the RC. See Figure 2 for examples of some confounding pathologies.

During the enrollment visit, participants were introduced to a HOCT device using printed material, without a physical device present in the clinic, and underwent imaging with an IO-OCT (Cirrus 5000, Carl Zeiss Meditech) with axial resolution of 5 µm and transverse resolution of 15 µm to establish a "baseline" status of the macula with 6 mm × 6 mm scans of both eyes. After verification of eligibility, participants were assigned a HOCT device that was delivered to participants' home by a courier service. Participants set up the device using the on-screen tutorial. To continue in the study, participants had to achieve successful initial HOCT automatic calibration in the study eye that was transmitted to the Notal Health Cloud. Continuing participants were instructed to self-image the study eye and, if eligible and successfully calibrated, also the fellow eye with the HOCT every day for 5 consecutive weeks, including on the days of scheduled office visits. Remote telephone technical support was made available by the physician-supervised Health Insurance Portability and Accountability Act-compliant digital health care provider. The Notal Vision Monitoring Center (NVMC) was available to participants by toll-free phone service. In-clinic study visits were scheduled for week 1 and week 5. To enrich the dataset with cases with borderline HRS, the daily HOCT scans of participants identified by the investigator as having absent HRS at enrollment were reviewed by the RC, and the RC triggered an alert for an interim clinic visit if any HRS was noted, including borderline HRS that were identified on 2 consecutive HOCT scans. In-office OCT images, best-corrected visual acuity, and any symptoms or adverse event (AE) information were collected at the prescheduled and interim visits. Since IO-OCT imaging is the most common method in many cases, the only method used for monitoring eyes with nAMD, a comparator of IO-OCT was accepted by the FDA.

The primary effectiveness endpoints were positive percent agreement (PPA) and negative percent agreement (NPA) of the HRS status (central 3 × 3 area) between RC-graded HOCT and IO-OCT scans, success rate of initial HOCT setup, and success rate of calibration and daily HOCT self-imaging attempts. In contrast to sensitivity and specificity, the endpoints of PPA and NPA were chosen because the clinical reference standard of IO-OCT is not considered an error-free reference gold standard, given the variation in human interpretation. The primary analysis, PPA and NPA for TRO visualization, was agreed upon by the FDA and prespecified in the study protocol and was based on repeated measurements of the primary eyes using the optimal weighting method to test H_0: $P \leq 0.8$ vs. H_a: $P > 0.8$ with normal approximation and a 2-sided significance level of 0.05.[21,22] The RC reviewed all the B-scans for the entire macula with both HOCT and IO-OCT. The primary analysis was performed using both a naïve estimate, which considered only IO-OCT scans that had a matching gradable HOCT scan, and a pragmatic estimate, which included IO-OCT scans that had no matching gradable HOCT scan.[23] To inform about the differences in manifestation of HRS between HOCT and IO-OCT, upon completion of the masked grading and locking of the primary endpoints, the RC performed an unmasked review of all cases of discrepancies in HRS status between the HOCT and IO-OCT, including a detailed side-by-side review and a summary of all the cases of disagreement. Secondary analyses of the PPA and NPA by type of HRS (SRO and IRO) and when using a 6 mm × 6 mm IO-OCT scan area comparator were performed. Positive predictive value (PPV), defined as the percent of eyes that have HRS present on the IO-OCT volume scans among the primary eyes that have HRS present on HOCT volume scan, and negative predictive value (NPV), defined as the percent of eyes that have HRS absent on the IO-OCT volume scans among the primary eyes that have HRS absent on HOCT volume scans, were derived. Positive predictive value and NPV provided insight into the accuracy of positive and negative test results in a given population, based on test accuracy and existing disease prevalence. Safety outcomes of interest were any AEs occurring during the conduct of the study.

Statistical analysis was performed using SAS version 9.4 (SAS Institute, Inc).

Results

A total of 198 participants signed informed consent and were enrolled. Of these, 7 (3.5%) were screen failures, 8

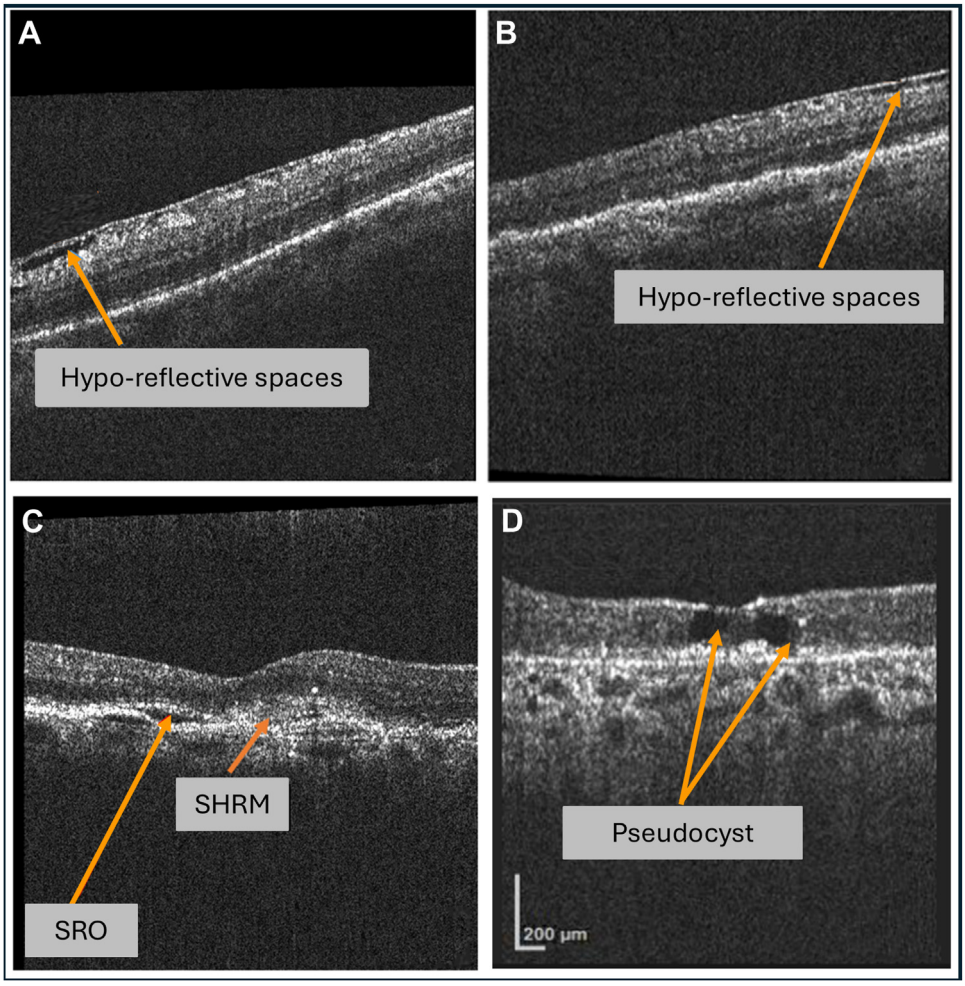


Figure 2. Examples of confounding pathologies over B-scans of the home OCT. **A, B,** Hyporeflective spaces underneath ERM. **C,** SRO and SHRM. **D,** Hyporeflective spaces representing pseudocysts seen as hyporeflective, circular-shaped lesions. ERM = epiretinal membrane; SHRM = subretinal hyperreflective material; SRO = subretinal hyporeflective spaces.

(4.0%) withdrew consent prior to initiating participation, 2 (1.0%) were excluded due to the inability to scan with the HOCT, and 1 (0.5%) was excluded because of an inability to return for follow-up. Therefore, 180 participants (90.9%) comprised the safety population. Within the safety population, 12 (6.7%) participants were excluded since they did not have an HOCT to pair with the IO-OCT at week 1, interim (if available), and week 5 visits. The remaining 168 participants (93.3%) formed the "visualization analysis population" (those who underwent HOCT imaging and clinic-based OCT at week 1, interim, or week 5 visits). From the visualization analysis population, another 8 (4.4%) participants were excluded due to the absence of a matching pair with the IO-OCT to form 160 participants (88.9%) in the "modified visualization analysis population" (mVAP). These details of the enrollment flow are presented in Figure 3. Primary analyses were based on mVAP.

The mean age of the safety population was 77.1 ± 7.2 years (range: 55–92 years). There was a slight preponderance of women (56.7%), and most subjects were Caucasian (96.7%) and non-Hispanic, non-Latinos (98.9%). There were 92 (51.1%) primary eyes and 35 (25.5%) secondary eyes with nonactive nAMD (HRS absent) and 88 (48.9%) primary eyes and 31 (22.6%) secondary eyes with active nAMD (HRS present). None of the primary eyes had early or intermediate age-related macular degeneration, while 61 (44.5%) of the secondary eyes had intermediate age-related macular degeneration and 10 (7.3%) had early age-related macular degeneration. The proportion of primary eyes that were pseudophakic at the start of the study was 63.9%. None of the eyes was treatment naïve, and the mean number of prior injections for nAMD was 26.4 ± 26.5 for the primary eyes and 12.1 ± 21.4 for the secondary eyes. The mean best-corrected visual acuity was 0.301 logarithm of the minimum angle of resolution (20/40) and 0.234 (20/34) for the primary and secondary eyes, respectively. On standard IO-OCT, SRO was present in 22.2%, IRO in 13.9%, both SRO and IRO in 5.6%, and absent (no SRO and IRO) in 58.3% of primary eyes per the principal investigator's assessment upon enrollment. The findings for mVAP were similar. Detailed demographics, racial distribution, and relevant baseline clinical characteristics are shown in

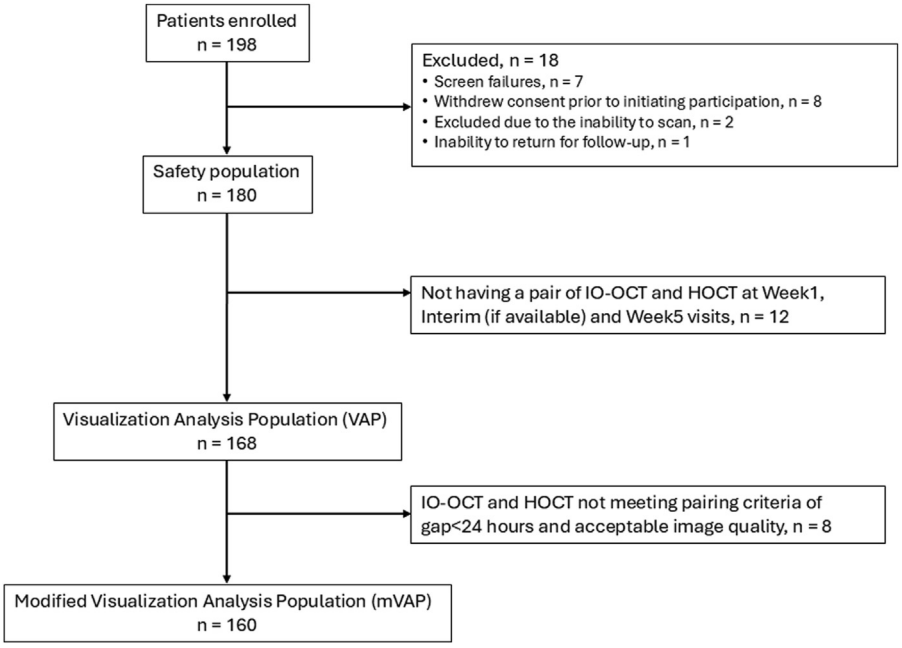


Figure 3. Subjects enrollment and analyses population diagram. HOCT = home OCT; IO-OCT = in-office OCT.

Tables 1 and 2. Table 3 details the concomitant pathologies identified in primary eyes by the RC on IO-OCT scans in the mVAP. The most common pathologies observed in the week 1 visit were pigment epithelial detachments (83.9%), hyperreflective retinal spots/foci (78.2%), epiretinal membrane (38.7%), pseudocysts (24.2%), subretinal pigment epithelium hyporeflective areas (21.0%), geographic atrophy (20.2%), and subretinal hyper-reflective material (15.3%). The findings from the interim and week 5 visits were similar.

For the safety population (N = 180), the success rate of HOCT self-imaging was 96.1% (95% confidence interval

Table 1. Demographics of Safety and Modified Visualization Analysis Populations

	Safety Population (n = 180)	mVAP (n = 160)
Age		
Mean ± SD	77.1 ± 7.2	76.8 ± 7.2
Median	77.5	77.0
Min, Max	55, 92	55, 92
Sex		
Male	78 (43.3%)	70 (43.8%)
Race		
Asian	1 (0.6%)	1 (0.6%)
Black or African American	4 (2.2%)	3 (1.9%)
White	174 (96.7%)	155 (96.9%)
Not reported	1 (0.6%)	1 (0.6%)
Ethnicity		
Not Hispanic or Latino	178 (98.9%)	158 (98.8%)
Not reported	2 (1.1%)	2 (1.3%)
Education		
Less than a high school degree	8 (4.4%)	7 (4.4%)
High school degree	43 (23.9%)	38 (23.8%)
Some college (no degree)	47 (26.1%)	44 (27.5%)
College degree (associate or bachelor's degree)	51 (28.3%)	45 (28.1%)
Graduate degree	25 (13.9%)	21 (13.1%)
Other*	6 (3.3%)	5 (3.1%)
Study eye		
Right eye	93 (51.7%)	82 (51.3%)

mVAP = modified visualization analysis population; SD = standard deviation.
*Including some graduate school, trade school, and tech school.

Table 2. Baseline Characteristics of Safety and mVAP

	Safety Population		mVAP	
	*Primary Eye** (N = 180)	*Secondary Eye*† (N = 137)	*Primary Eye* (N = 160)	*Secondary Eye* (N = 123)
AMD diagnosis‡				
Early AMD	0 (0.0%)	10 (7.3%)	0 (0.0%)	9 (7.3%)
Inactive nAMD - (no HRS)	92 (51.1%)	35 (25.5%)	84 (52.5%)	27 (22.0%)
Intermediate AMD	0 (0.0%)	61 (44.5%)	0 (0.0%)	58 (47.2%)
Active nAMD - (HRS present)	88 (48.9%)	31 (22.6%)	76 (47.5%)	29 (23.6%)
Lens status				
Phakic (cataract present)	65 (36.1%)	47 (34.3%)	61 (38.1%)	46 (37.4%)
Pseudophakia	115 (63.9%)	90 (65.7%)	99 (61.9%)	77 (62.6%)
Ocular media assessment				
Main vessels and the small vessels are clearly seen	179 (99.4%)	136 (99.3%)	159 (99.4%)	122 (99.2%)
Both main and small vessels cannot be seen	1 (0.6%)	1 (0.7%)	1 (0.6%)	1 (0.8%)
Visual distortions				
Present	17 (9.4%)	7 (5.1%)	15 (9.4%)	7 (5.7%)
Absent	163 (90.6%)	130 (94.9%)	145 (90.6%)	116 (94.3%)
Blurry vision				
Present	57 (31.7%)	28 (20.4%)	54 (33.8%)	26 (21.1%)
Absent	123 (68.3%)	109 (79.6%)	106 (66.3%)	97 (78.9%)
Scotoma				
Present	57 (31.7%)	28 (20.4%)	54 (33.8%)	26 (21.1%)
Absent	123 (68.3%)	109 (79.6%)	106 (66.3%)	97 (78.9%)
Prior injections				
Mean ± SD	26.4 ± 26.5	12.1 ± 21.4	25.3 ± 26.6	12.0 ± 21.9
Median	17.0	0.0	16.0	0.0
Min, Max	0, 128	0, 125	0, 128	0, 125
Spherical equivalent (diopters)				
Mean ± SD	0.066 ± 1.924	0.044 ± 1.810	0.109 ± 1.881	0.135 ± 1.672
Best-corrected visual acuity				
Mean logMAR (Snellen)	0.301 (20/40.0)	0.234 (20/34.3)	0.281 (20/38.2)	0.211 (20/32.5)
SD logMAR	0.251	0.324	0.226	0.309
20/40 or better	110 (61.1%)	106 (77.4%)	102 (63.8%)	100 (81.3%)
20/41–20/80	50 (27.8%)	20 (14.6%)	44 (27.5%)	16 (13.0%)
20/81–20/200	14 (7.8%)	9 (6.6%)	12 (7.5%)	6 (4.9%)
20/201–20/320	6 (3.3%)	1 (0.7%)	2 (1.3%)	1 (0.8%)
Worse than 20/320	0 (0.0%)	1 (0.7%)	0 (0.0%)	0 (0.0%)
PI assessment of retinal HRS on IO-OCT‡				
Both IRF and SRF	10 (5.6%)	3 (2.2%)	8 (5.0%)	3 (2.4%)
SRF only	40 (22.2%)	13 (9.5%)	36 (22.5%)	13 (10.6%)
IRF only	25 (13.9%)	14 (10.2%)	20 (12.5%)	11 (8.9%)
No IRF nor SRF	105 (58.3%)	107 (78.1%)	96 (60.0%)	96 (78.0%)

AMD = age-related macular degeneration; HRS = hyporeflective spaces; IO-OCT = in-office OCT; IRF = intraretinal fluid; logMAR = logarithm of the minimum angle of resolution; mVAP = modified visualization analysis population; nAMD = neovascular age-related macular degeneration; SD = standard deviation; SRF = subretinal fluid.
*Primary eye = study eye.
†Secondary eye = the fellow eye of the study eye.
‡AMD diagnosis based on medical records. OCT HRS status based on investigator review of IO-OCT taken at enrollment visit.

[CI]: 92.2%–98.4%). One hundred twenty patients (66.7%) contacted the NVMC for technical support. Thirty-four patients (18.9%) called for support in setting up the device. The NVMC contacted 47 (26.1%) with low adherence (no device use for 2 consecutive days), a rate of 0.21 calls per user per month, to remind them to perform self-imaging. The total number of self-images of the primary and secondary eyes during the 5 weeks of the study were 5426 and 4012, respectively. The mean (standard deviation) MSI results of the primary and secondary eyes were 4.40 (1.26) and 4.58 (1.28), respectively, with 97.5% of scan with MSI ≥2.

Table 4 shows the results of the primary analysis of mVAP, PPA, and NPA for the target condition of HRS visualization between the index test of HOCT and same-day comparator IO-OCT using a 3 × 3 area and Tables 5 and 6 include the results of the secondary analyses of PPA and NPA for HRS visualization by HRS subtype of SRO and IRO. A post hoc, unmasked, side-by-side analysis of the 39 discrepancies identified in the primary results was performed by the RC. Disagreement involved SRO in 30 (77%) cases, IRO in 12 (31%) cases, and both in 2 (5%) cases. Reasons for disagreements were determined to be grader error in 21 (54%) cases, borderline HRS on 1 device

Table 3. Concomitant Pathologies Identified by the Reading Center on In-Office OCT Scans in the Modified Visualization Analysis Populations

	Week 1 (N = 124)*	Interim (N = 47)	Week 5 (N = 138)
Any pathology†	122 (98.4%)	47 (100%)	134 (97.1%)
Epiretinal membrane	48 (38.7%)	15 (31.9%)	47 (34.1%)
Macular hole	0 (0.0%)	1 (2.1%)	1 (0.7%)
Pseudocysts	30 (24.2%)	12 (25.5%)	30 (21.7%)
Outer retinal tubulations	6 (4.8%)	2 (4.3%)	1 (0.7%)
Hemorrhage	0 (0.0%)	1 (2.1%)	0 (0.0%)
Pigment epithelial detachments	104 (83.9%)	42 (89.4%)	112 (81.2%)
Subretinal hyper-reflective material	19 (15.3%)	3 (6.4%)	18 (13.0%)
Sub-RPE hyporeflective areas	26 (21.0%)	17 (36.2%)	28 (20.3%)
Geographic atrophy	25 (20.2%)	15 (31.9%)	33 (23.9%)
Hyper-reflective retinal spots	97 (78.2%)	41 (87.2%)	120 (87.0%)
Not graded - poor image quality	1 (0.8%)	0	0

RPE = retinal pigment epithelium.
*N = number of eyes with available data.
†Only the scans of the primary eye were graded.

that was missed on the other was identified in 21 (54%) cases, relatively superior image quality on IO-OCT in 5 (13%) cases and HOCT in 2 (5%) cases, and poor image quality on both devices in 1 (3%) case. A complete analysis of all cases of discrepancies is included in Table S5 (available at www.ophthalmologyscience.org). Table S6 (available at www.ophthalmologyscience.org) compares agreement between HOCT and 6 mm × 6 mm (6 × 6) IO-OCT in the mVAP population. Utilizing the wider IO-OCT scan area, 9 eyes moved from "IO-OCT = no TRO" to "IO-OCT = TRO." Of these, 1 eye was graded with HRS on HOCT, indicating HRS on the edges of the 3 × 3 area that is also seen on IO-OCT, and in 8 eyes, HRS was seen on IO-OCT 6 × 6 only, indicating the location of HRS outside of the 3 × 3 area. Overall, HRS were seen outside the 3 × 3 area in only 4.7% of cases, supporting the assumption that 3 × 3 scan area is adequate for capturing TRO in nAMD eyes. Positive predictive value for HOCT ranged from 0.849 (95% CI: 0.786–0.913) to 0.877 (95% CI: 0.818–0.936) based on 163 pairs of scans. Negative predictive value ranged from 0.861 (95% CI: 0.803–0.920) to 0.868 (95% CI: 0.808–0.929) based on 144 pairs of scans. Table S7 (available at www.ophthalmologyscience.org) summarizes PPV and NPV for all HRS subtypes using the naïve and pragmatic approaches for both 3 × 3 and 6 × 6 IO-OCT scan areas. Fifty-one interim visits were conducted in response to the daily RC review for 51 (28.3%) participants. In 32 of these visits, a new HRS was identified by the RC on the HOCT scans, and of these, 28 (87.5%) were also identified on the IO-OCT. Ten AEs were reported for 6 participants. Of these, 4 were considered serious AEs. All serious AEs were nonocular in nature (myocardial infarction, pulmonary edema, fall, and COVID-19 infection). Two AEs in 2 participants were ocular in nature (ocular pain and redness at the intravitreal injection site and eyelid hordeolum). No ocular AEs involving vision loss were reported. None of the AEs were considered

Table 4. Visualization Study—Primary Analysis of PPA and NPA for TRO

	In-Office OCT (3 × 3 mm Area) vs. HOCT	
	PPA Scans (105 Eyes)	*NPA Scans (95 Eyes)*
In-office OCT scans	163	146
HOCT scans with TRO	143	20
HOCT scans without TRO	19	124
HOCT scans not gradable	1	2

	PPA (95% CI)	*P* Value	NPA (95% CI)	*P* Value
Naïve estimation*	0.866 (0.804–0.928)	0.036	0.861 (0.804–0.918)	0.037
Pragmatic estimation	0.864 (0.802–0.926)	0.043	0.849 (0.792–0.907)	0.094

CI = confidence interval; HOCT = home OCT; NPA = negative percent agreement; PPA = positive percent agreement; TRO = total retinal hyporeflective spaces.
*The analysis was performed using both a naïve estimate, which considered only in-office OCT scans that had a matching gradable home OCT scan, and a pragmatic estimate, which included in-office OCT scans that had no matching gradable home OCT scan.

Table 5. Visualization Study—Secondary Analysis of PPA and NPA for SRO

	In-Office OCT (3 × 3 mm Area) vs. HOCT	
	PPA Scans (82 Eyes)	*NPA Scans (118 Eyes)*
In-office OCT scans	117	192
HOCT scans with SRO	100	22
HOCT scans without SRO	16	168
HOCT scans not gradable	1	2

	PPA (95% CI)	NPA (95% CI)
Naïve estimation*	0.846 (0.770–0.922)	0.883 (0.835–0.931)
Pragmatic estimation	0.836 (0.759–0.914)	0.874 (0.826–0.923)

CI = confidence interval; HOCT = home OCT; NPA = negative percent agreement; PPA = positive percent agreement; SRO = subretinal hyporeflective spaces.
*The analysis was performed using both a naïve estimate, which considered only in-office OCT scans that had a matching gradable home OCT scan, and a pragmatic estimate, which included in-office OCT scans that had no matching gradable home OCT scan.

related to the HOCT device, and all AEs resolved before study completion.

Discussion

In the visualization study, a total of 180 of the 198 enrolled subjects (90.9%) received an HOCT device to their home, and 96.1% of them successfully calibrated and self-imaged. The study population was reasonably representative of the intended target population for the HOCT system in terms of age, gender, ethnicity, education level, visual acuity, status of the fellow eye, and other existing ocular pathologies. Concomitant pathologies identified by the RC were present in almost all eyes, supporting the robustness of the findings. The subjects demonstrated a very high success rate in device receipt at home, guided setup and calibration using a nonsupervised video tutorial, and initiation of self-imaging. Over the 5-week study period, 9438 scans with acceptable image quality were acquired and transmitted to the NVMC. The high success rate of HOCT self-imaging met the primary endpoint and demonstrated the feasibility of OCT imaging at home.

The primary efficacy endpoints for the evaluation of the performance of the index test were defined as PPA and NPA of TRO in the 3 × 3 area. The use of PPA and NPA instead of sensitivity and specificity was due to the comparator of IO-OCT; however, it provided a practical context to the results and was acceptable to the FDA. Using the naïve estimation approach and repeated measurements, the PPA of 0.866 (95% CI: 0.804–0.928; $P = 0.036$) and NPA of 0.861 (95% CI: 0.804–0.918; $P = 0.037$) met the prespecified performance goals for the study's primary outcome as agreed with the FDA. Two analytical approaches discussed by Begg et al[23] were utilized. The naïve estimation is considered to be more reliable for such an analysis since the proportion of ungradable scans was extremely low (3/335 = 0.9%) and the 3 ungradable scans included a case when HRS was present on IO-OCT and 2 cases when HRS was absent on IO-OCT, indicating that an ungradable scan was not associated with a specific HRS status. Moreover, all eyes with an ungradable scan had a gradable scan at another time point, indicating that a status of "ungradable" was not a characteristic specific to certain eyes. Using a pragmatic estimation, the PPA was 0.864 (95% CI: 0.802–0.926; $P = 0.043$), and the NPA was 0.849 (95% CI: 0.792–0.907; $P = 0.094$). Although the lower 95% CI (0.792) for NPA using a pragmatic estimation was 0.008 lower than the performance goal of 0.8, the point estimation of 0.849 was 0.05 higher than the performance goal, supporting the conclusion that the agreement in visualization of HRS was very similar by both analytical approaches. There were 39 pairs of images where there were discrepancies between the HOCT and IO-OCTs in the grading of the presence of SRO, IRO, or SRO and IRO in the study. In most cases, the HRS were subtle or only questionably present, and graders were often discrepant even within the same scan type. Discrepancies of this nature are expected when findings are borderline with subjective interpretation by human graders. Given the low overall proportion of discrepancies, this type of variability in the study was consistent with typical RC performance when grading eyes with nAMD.[24] There was a slight tendency for human readers to falsely grade SRO as being present on HOCT images, likely due to subtle differences in signal reflectivity between HOCT and IO-OCT in the outer retinal bands associated with questionable SRO presence, and further exploration of that assumption may be of interest. Otherwise, there were no systematic differences in grading between the 2 instruments. Overall, image quality and retinal HRS detection by human readers were similar between HOCT and IO-OCT devices.

Table 6. Visualization Study—Secondary Analysis of PPA and NPA for IRO

	In-Office OCT (3 × 3 mm Area) vs. HOCT	
	PPA Scans (50 Eyes)	*NPA Scans (131 Eyes)*
In-office OCT scans	75	234
HOCT scans with IRO	100	22
HOCT scans without IRO	16	168
HOCT scans not gradable	1	2

	PPA (95% CI)	NPA (95% CI)
Naïve estimation*	0.754 (0.637–0.872)	0.963 (0.934–0.992)
Pragmatic estimation	0.753 (0.635–0.871)	0.957 (0.927–0.987)

CI = confidence interval; HOCT = home OCT; IRO = intraretinal hyporeflective spaces; NPA = negative percent agreement; PPA = positive percent agreement.
*The analysis was performed using both a naïve estimate, which considered only in-office OCT scans that had a matching gradable home OCT scan, and a pragmatic estimate, which included in-office OCT scans that had no matching gradable home OCT scan.

Ancillary clinical outcomes were assessed, including PPA and NPA for SRO and IRO HRS subtypes, as well as a comparison of all subtypes of HRS in the central 3 × 3 on the HOCT volume scan with the central 6 × 6 of the IO-OCT volume scan. Results were acceptable based on the context of use for HOCT. A small proportion of HRS (4.7%) of any type were exclusively identified outside of the 3 × 3 scan area. These pose a more limited risk to vision due to the distance from the critical foveal region. Given the typical nAMD disease progression dynamics, it is expected that HRS would ultimately be observed in the 3 × 3 scan area with serial HOCT scanning.

Positive predictive value is a useful and slightly different method for assessing the performance of a diagnostic test, which takes into account the prevalence of the finding of interest in a population. As the prevalence of disease increases (i.e., true positives are more common), the likelihood of a false positive decreases. Therefore, predictive value can change over time or in different places, while sensitivity and specificity do not change, as these are characteristics of the test itself. The results of PPV and NPV are similar to the PPA and NPA results, illustrating that the prevalence of active disease in the study population was acceptable, with approximately half of the eyes with TRO present and half of the eyes with TRO absent at any given time during an office visit. Positive predictive values involving TRO were very similar to PPVs using the IO-OCT 6 mm × 6 mm area, while NPVs were slightly lower involving TRO visualized using the IO-OCT 6 mm × 6 mm area. During the study, the subjects returned to the office at week 1 and week 5, and if HRS was identified by the RC in eyes that enrolled without HRS in 2 consecutive days, they returned for an interim visit. No treatment decisions were dictated by the study protocol, and they were at the discretion of the investigator. Approximately half of the 51 interim visits led to a decision to treat. In the other half, treatment decision-making was not impacted by HOCT findings as the extent of HRS, identified by the RC, was not deemed treatment relevant, the interval from the last treatment was too short to allow treatment, or a treatment was already scheduled for the upcoming week 5 visit.

Based on the study's findings, the following note was included in the product labeling to mitigate misinterpretation of the true HRS status: scans with poor image quality below MSI of 2 may be unreliable. Taking into account this labeling instruction, the element of the physician report of raw B-scans, with indication of image quality, is properly validated.

In conclusion, findings of the study support the use of the patient self-operated HOCT system as a reliable tool to visualize retinal HRS from serial spectral-domain OCT scans while monitoring at home between routine clinical visits as part of the overall intended use.

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The Home OCT Pivotal Studies Research Group is listed online in Appendix 1 (available at www.ophthalmologyscience.org).

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Abbreviations and Acronyms:
AEs = adverse events; **CI** = confidence interval; **FDA** = Food and Drug Adminisration; **HOCT** = home OCT; **HRS** = hyporeflective spaces; **IO-OCT** = in-office OCT; **IRO** = intraretinal hyporeflective spaces; **MSI** = manufacturer signal quality index; **mVAP** = modified visualization analysis population; **nAMD** = neovascular age-related macular degeneration; **NPA** = negative percent agreement; **NPV** = negative predictive value; **NVMC** = Notal Vision Monitoring Center; **PPA** = positive percent agreement; **PPV** = positive predictive value; **RC** = reading center; **SRO** = subretinal hyporeflective spaces; **TRO** = total retinal hyporeflective spaces.

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